Paper 3

1 (a) time = $\dfrac{\text{distance}}{\text{speed}} = \dfrac{0.15 \times 10^{12}}{3.00 \times 10^{8}}$

$= 8.3$ min A1

Comments: never leave answer in fraction form.

(b) units of k = N m^{-1} = kg s^{-2}, units of T = s, units of m = kg

$C = T\sqrt{\dfrac{k}{m}}$

units of $C = \text{s}\sqrt{\dfrac{\text{kg s}^{-2}}{\text{kg}}} = 1$ A1

Hence C has no units. A0

Comments: Presentation of answer is important as it is a 'show' type of question. Often mistakes were made by equating quantity with unit e.g. k = kg s^{-2}

(c) (i) $C = T\sqrt{\dfrac{k}{m}} = 0.242\sqrt{\dfrac{239}{0.300}} = 6.83$ A1

(ii) $\dfrac{\Delta C}{C} = \dfrac{\Delta T}{T} + \left(\dfrac{1}{2}\right)\dfrac{\Delta k}{k} + \left(\dfrac{1}{2}\right)\dfrac{\Delta m}{m}$

$= \dfrac{1}{100} + \left(\dfrac{1}{2}\right)\dfrac{3}{100} + \left(\dfrac{1}{2}\right)\dfrac{2}{100}$

$= 0.035$ C1

$\Delta C = 0.2$

$C = (6.8 \pm 0.2)$ A1

(iii) accuracy: determined by the closeness of the value(s)/ measurement(s) to the true value B1

precision: determined by the range of values/measurements B1

2 (a) Straight line graph through origin, so A1